PAPER_923: SCm Phonon Resonance Acceleration a_res

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-11 **Session:** 211 **Source:** SCm phonon gap implementation (scm_phonon_resonance.py) **Calculator:** SCmPhononResonanceAccelerationCalc (CP4 #507) **CVW:** v2.0.0 compliant

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Abstract

Derives the SCm phonon resonance acceleration a_res = (F_{U,Bi}/F_U) * Phi_{1.25THz}(omega) * S_26([SSq]), the quantitative acceleration produced when UQFF vacuum buoyancy couples to the 1.25 THz palladium-deuterium lattice phonon resonance. At resonance (omega = omega_SCm), Phi_{1.25THz} reaches its peak value ~10^20 and a_res enters the phonon-dominated regime (~10^20 m/s^2), representing the strongest gravitational modification in the UQFF framework. The 6-class module covers resonance acceleration, linewidth gamma-sweeps, vacuum density coupling, frequency scans, phonon damping evolution, and multi-layer phonon-gravity coupling across all 26 UQFF layers.

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1. Core Equations

Section A: Lagrangian

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L_phonon = a_res * V_region * S_26
a_res = (F_{U,Bi}/F_U) * Phi_{1.25THz}(omega) * S_26([SSq])
Phi_{1.25THz}(omega) = Phi_0 * exp[-(omega - omega_SCm)^2 / (2 * Gamma^2)]
S_26 = sum_{k=1}^{26} exp(-[SSq] * k / 26)
```

Section B: VDS/DVP/BH Number Systems

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VDS: rho_vac(omega) = rho_0 * Phi_{1.25THz}(omega) / Phi_0
DVP: p_n = Product_{k=1}^{n} (1 + a_res_k / g_Newton)  (n = 1..26)
BH: h_B = sum_{n=1}^{26} cos(2*pi*n*omega/omega_SCm) * Phi_n
```

Section SM: SM Anchors

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omega_SCm = 2*pi * 1.25e12 rad/s  (Pd-D lattice phonon)
[SSq] = 0.57  (calibrated UQFF coupling)
Phi_0 = 10^20  (peak phonon amplitude)
Gamma_default = 2*pi * 0.1e12  (linewidth)
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2. Parameters

Parameter	Default	Description
F_UBi	0.6 N	Buoyancy force
F_U	1.0 N	Unified force
omega	omega_SCm	Angular frequency
Gamma	2*pi*0.1e12	Linewidth
Phi_0	10^20	Peak phonon amplitude

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3. Key Results

Configuration	a_res (m/s^2)	Regime
On-resonance (F_UBi/F_U=0.6)	~6.0e19	phonon-dominated
Half-ratio (F_UBi/F_U=0.3)	~3.0e19	phonon-dominated
Off-resonance (omega=0.5 THz)	~0	sub-resonant

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4. Physical Interpretation

The resonance acceleration a_res quantifies the extreme gravitational modification possible when vacuum phonon modes align with the 1.25 THz Pd-D lattice resonance. This provides the mechanism for LENR excess heat via buoyancy-mediated phonon-gravity coupling, and explains why specific lattice frequencies (palladium-deuterium, nickel-hydrogen) produce anomalous energy output. The 26-layer polylog structure ensures the acceleration is distributed across all UQFF vacuum strata, preventing single-layer divergence.

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5. References

- PAPER_917: Exponential strain phonon time-evolution
- PAPER_919: Sgr A* flare contrast vs Gamma
- scm_phonon_resonance.py: 6-class standalone module
- et_phonon_resonance.py: ResonanceAccelerationTerm (Section 4b)